

Arithmetic SOS for the truncated Weil form: what exists, what the smallest window shows, what would be a factor

D. Joubert

4 September 2026

Abstract

On a window where $Q_L \succ 0$, a sum-of-squares factorization exists: $Q_L = \|Q_L^{1/2} \cdot\|^2$. That factor is spectral (or Cholesky). It is not built from the primes. An *arithmetic* SOS is a factor T_L written from P , A and the towers T_p only, with $T_L^* T_L = Q_L$. At $\mu = 3$ — one prime, one negative mode of $P + A$ — the Cholesky factor on V_9 is dense, T_2 is indefinite, and completing the square against the lag at $\log 2$ does not produce an explicit T_L . The closed SOS that does exist is the zero-side Gram. An arithmetic SOS for every L is point 4 of the notebook: RH.

1 Three factorizations, only one arithmetic in intent

On V_N at a window with $Q \succ 0$,

$$Q = \sum_{j=1}^N \lambda_j v_j v_j^\top = C^\top C = \sum_{\gamma} \hat{\eta}(\gamma) \hat{\eta}(\gamma)^\top.$$

The first two are $Q^{1/2}$ (spectral / Cholesky). They were computed at $\mu = 11$ (47 squares, certified by congruence, not by ball Cholesky). The third is $T_z f = (\hat{f}(\gamma))_\gamma$, closed form, and uses the zeros. An arithmetic T_L would satisfy $T_L^* T_L = P + A - \sum T_p$ with each row of T_L a local pairing (archimedean kernel or lag $\log p^k$), no γ and no eigenvector of Q .

2 The smallest laboratory: $\mu = 3$

Only the tower of 2 is present. Spectro on V_9 :

$$\lambda_{\min}(P + A) = -0.073, \quad \text{spec}(T_2) \subset [-0.490, 0.490], \quad \lambda_{\min}(Q) = 1.03 \times 10^{-7}.$$

$P + A$ has *one* negative direction. T_2 is indefinite of half-dimension (four negative eigenvalues on V_9). Q is positive with the known razor. The numerical Cholesky factor C of Q on V_9 is dense in the lower triangle: no band, no visible Toeplitz, no displacement rank that would write C as a short combination of the $\Theta(\log 2)$ matrix and the archimedean kernel.

Completing the square fails to be explicit. Write $P + A = A_+ - \lambda_- u u^\top$ with $\lambda_- = 0.073$. Then

$$Q = A_+ - \lambda_- u u^\top - T_2.$$

This is a difference of an PSD piece and an indefinite piece. It is not A_+ plus a square built from $U_{\log 2}$, because T_2 itself is not a square: $\Theta_f(\log 2) = \langle f, \tau_{\log 2} f \rangle$ is a correlation, not a norm. Expanding $\tau_{\log 2}$ in the cosine basis recovers the matrix of T_2 , not a list of arithmetic linear forms whose Gram is Q .

3 What an arithmetic T_L would look like

A candidate with the right shape, not known to work:

$$(T_L f)_p = \sqrt{\log p} p^{-1/4} (\alpha_p f + \beta_p \tau_{\log p} f) \quad \text{plus an archimedean row } T_\infty f.$$

Then $\|T_L f\|^2$ produces $\Theta(\log p)$ cross terms and $\|f\|^2, \|\tau f\|^2$ squares. Matching the coefficients of $P + A - \sum T_p$ is a finite linear algebra problem on each window — and a moving target as L grows (new primes, new lags $k \log p$). At $\mu = 3$ this ansatz has four real scalars (α_2, β_2) plus the archimedean row. It does not have enough parameters to reproduce a 9×9 PSD matrix of rank 9. Adding internal lags $k = 2, 3, \dots$ does not help: $2^k \leq 3$ forbids them.

The zero-side factor has as many rows as there are γ in band, which is the right count. The Euler side has as many rows as there are prime powers $\leq \mu$, which is too few (four towers at $\mu = 11$, against 47 dimensions). An arithmetic SOS must therefore use a *continuous* archimedean fibre, not only the discrete lags. That fibre is the Weil archimedean integral, which is not itself a square for $L > 0.85$. The mixing of a non-square continuous piece with four discrete lags is the whole problem.

Status

Explored, not constructed. On every finite V_N with $Q \succ 0$ an SOS exists and is $Q^{1/2}$. The Cholesky of the smallest PSD window is unstructured. The Euler lags are too few to span the factor. The closed SOS is the list of evaluators at the zeros. Passing from that list to a factor written from P, A, T_p for every L is RH.